

Systems for Design and Implementation : Assignment 3

BRONZE CHALLENGE IS NOT COMPLETE WITHOUT:

- unit tests and coverage for all the important CRUD operations
- thorough testing has to be ensured for the entire DATABASE

SILVER CHALLENGE

- first design and then implement the database tables and relationships necessary for the USER, ROLES and PERMISSIONS
- implement fullstack login behaviour for the different types of users:

- admin [FULL PERMISSIONS]

- normal user [RESTRICTED PERMISSIONS]

NOTE: no need for all the aspects of AUTHENTICATION and AUTHORIZATION to be implemented this will be required for the next assignment, for now just implement the

DATABASE INFRASTRUCTURE FOR ROLE AND PERMISSION MANAGEMENT.

it is not necessary that the data sent via network is encrypted and tokens need not be employed yet we focus only the PERSISTENCY aspects for now

- implement a **Real-Time Chat** feature using a **NoSql database** and **WebSockets** such that the users of your app can chat while on the web application. show real-time chatting with at least two separate logged in users

GOLD CHALLENGE

BE AMBITIOUS! Build an entire DATABASE INFRASTRUCTURE for a LOGGING SYSTEM on the BACK END of your WEB SERVER.

BE ALERT! Each ACTION performed by a LOGGED-IN USER needs to be persisted inside a LOG which will be implemented using tables in the database.

LOG ENTRY information should contain but not be restricted to:

USER_ID:GROUP_ID[ADMIN/USER]:ACTION_INFORMATION:TIMESTAMP

BE STEALTH! IMPLEMENT THE ENTIRE LOGICS OF A MECHANISM THAT DETECTS MALVOLENT BEHAVIOUR BASED ON USER ACTIONS AND PLACES SUSPICIOUS USERS IN AN OBSERVATION LIST THAT CAN BE VIEWED BY THE ADMIN ON THE FRONT END

Deadline for the bronze[5p] challenge: Week 10

Deadline for the silver[+2p] and gold[+3p] challenges without delay: Week 10[max grade 10]

Deadline for the silver[+1p] and gold[+1.5p] challenges with one delay: Week 11[max grade 7.5]

Deadline for the silver[+0.5p] and gold[+0.75p] challenges with two delays: Week 12[max grade 6.25]